

Lecture Notes 5a

Labor Supply and Unemployment

Intermediate Macroeconomics, EC2211

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Contents and literature

- Labor supply
- The labor market
- Unemployment

Literature:

- Jones (2024), chapter 7 and appendix to ch. 15
- *Steinsson (2025b), "[Work and leisure](#)", sections 1-2 and 4-4.4

Labor supply

How should we model purposeful choices by people?

- We often think of households as the decision-making unit
- We assume that households have **preferences**
- We represent preferences by a **utility function**
- We assume that households seek to maximize utility
- ... subject to resource or budget constraints

Questions

- Effect of wages or taxes on incentives to work?
- Effect of long-run growth on leisure?
- Effect of technology (machines $\rightarrow$ robots $\rightarrow$ AI) on labor demand?

Wages and labor supply

Consider the following (hypothetical!) offer:

I need research assistants the coming two weeks. You can volunteer to work H_1 hours the first week and H_2 hours the second week.

- The first week, I offer an hourly wage of EUR 15
- The second week, I offer an hourly wage of EUR 50

Does the wage affect how much you want to work?

Wages and labor supply

Consider now these two scenarios:

1. Your hourly wage will be of EUR 15 for the rest of your life
2. Your hourly wage will be of EUR 50 for the rest of your life

In both scenarios you can choose how many hours to work per week

Do you want to work more or less in scenario 1 or 2?

Keynes (1930), "Economic possibilities for our grandchildren"



John Maynard Keynes
1883-1946

- "What can we reasonably expect the level of our economic life to be a hundred years hence?"
- "I would predict that the standard of life in progressive countries one hundred years hence will be between four and eight times as high as it is to-day"
- "Mankind will be deprived of its traditional purpose [the struggle for subsistence]"
- "Must we not expect a general nervous breakdown?"
- "Three-hour shifts or a fifteen-hour week may put off the problem for a great while"

Hours worked 1870-1998

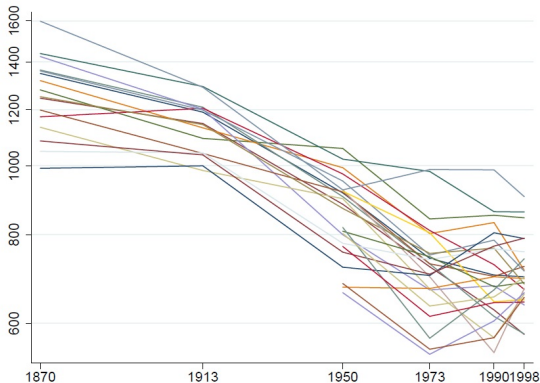


Figure 1: Average yearly hours worked per capita 1870–1998

Source: Maddison (2001). The sample includes the following 25 countries: Austria, Belgium, Denmark, Finland, France, Germany, Italy, Netherlands, Norway, Sweden, Switzerland, United Kingdom, Ireland, Spain, Australia, Canada, United States, Argentina, Brazil, Chile, Colombia, Mexico, Peru, Venezuela, Japan. Regressing the log of hours worked on a country fixed effect and year gives a slope coefficient of -0.00462 in the full sample (and -0.00398 for the period 1950–1998). Huberman and Minns (2007) provide similar data.

Hours worked in the U.S. 1950-2020

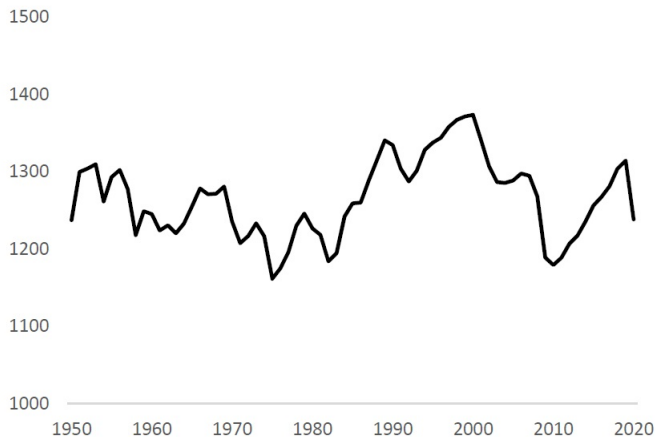


Figure 4: Annual Hours Worked per Person in the United States

Household preferences

Suppose that the household's utility function is:

$$U(C) - V(H) \tag{1}$$

where:

- $U(C)$ is utility from consumption of goods and services, C
- $V(H)$ is disutility from working and H denotes hours worked

Budget constraint

In this section, we focus on one-period problems and ignore savings etc.

Suppose that the household's budget constraint is:

$$C = wH + T \quad (2)$$

where:

- The left-hand side is money spent on consumption of goods and services
- The right-hand side is income:
 - wH , wage times hours worked, is labor income
 - T is other income, such as transfers

Household optimization

The household chooses consumption and labor to maximize utility subject to the budget constraint:

$$\max_{C,H} = U(C) - V(H) \quad (3)$$

subject to $C = wH + T$

Use the budget constraint to substitute for C in the utility function. The problem is then

$$\max_H = U(wH + T) - V(H) \quad (4)$$

Solving for the optimal labor-leisure choice

The first-order condition with respect to H is:

$$wU'(wH + T) - V'(H) = 0 \quad (5)$$

or if we use the budget constraint $C = wH + T$:

$$wU'(C) = V'(H) \quad (6)$$

Interpreting the solution

$$wU'(C) = V'(H)$$

- The left-hand side (LHS) shows the value of working one extra hour: we get w extra income to spend on consumption. $U'(C)$ is the extra utility from one unit of consumption.
- The right-hand side (RHS) shows the disutility from that extra hour of work
- If the LHS had been greater than the RHS, we should have chosen to work more, etc.

How is labor supply affected by the wage?

Suppose that preferences are

$$U(C) = \alpha \ln C \quad \text{and} \quad V(H) = -(1 - \alpha) \ln(1 - H)$$

Here the household's total time available has been normalized to unity so that $1 - H$ is leisure

Note that

$$U'(C) = \frac{\alpha}{C} \quad \text{and} \quad V'(H) = \frac{1 - \alpha}{1 - H}$$

Suppose also that $T = 0$. We then see from the first-order condition that the solution is

$$wU'(C) = V'(H) \implies H = \alpha \text{ and } C = \alpha w$$

How is labor supply affected by the wage?

In this particular setting, we found that $H = \alpha$. Labor supply does not depend on the wage! Why???

There is both a **substitution effect** and an **income effect** at work.

- Substitution effect: A higher wage raises the return to work. We want to work more.
- Income effect: A higher wage means that the same labor effort results in higher income. With higher income we want to "spend" more on both consumption and leisure.

The logarithmic preferences that we assumed (and which are often used in economics) have the special property that substitution and income effects are equally large.

How is labor supply affected by an income tax?

Consider an income tax τ . The household's budget constraint is now:

$$C = (1 - \tau)wH + T$$

We consider two cases:

- a. Tax revenue is vusted, $T = 0$
- b. Tax revenue is returned to households as lump-sum transfers¹, $T = \tau wH$

Do the implications for labor supply differ between the two cases?

¹Note that this is not internalized by the household in its decision problem since an individual household does not affect the size of the transfer

How is labor supply affected by an income tax?

Yes, the implications do differ!

We have already analyzed case (a) where $T = 0$. The income tax effectively reduces the wage:

- Because of the substitution effect, the household wants to work less.
- But the household also has less income and therefore wants to spend less on both consumption and leisure. Because of the income effect, the household wants to work more.
- The substitution and income effects cancel, and the tax has no effect on labor supply.

In case (b) however, labor supply falls.

- If H were unaffected, so would the household's income. There would then be no income effect.
- Clearly, the substitution effect will dominate here.

Why do Europeans work so little relative to Americans?

TABLE 7.2

Hours Worked per Person (U.S. = 100 in 2019)

	1960	1990	2019
United States	88	103	100
Italy	107	87	86
France	111	79	75
Germany	131	93	88
U.K.	114	96	96
Japan	130	125	110
South Korea	89	134	122

Source: Penn World Table, Version 10.0.

Why do Europeans work so little relative to Americans?

- Tax rates?
- Different preferences?
- Social multipliers?
- Labor regulation and unions?

Tax rates?

- Differences in tax systems, marginal tax rates in particular, key driver of differences in labor supply²
- Similar results found for Sweden³
- But: the empirical evidence reveals a mis-match between micro and macro estimates

²Prescott, E. (2004), "Why do Americans Work so Much More than Europeans?", *Federal Reserve Bank of Minneapolis Quarterly Review*, 28(1), pp. 2-13

³Olovsson, C. (2009), "Why do Europeans Work so Little?", *International Economic Review*, 50(1), pp. 39-61

Different preferences and social multipliers?

- Differences in preferences: do Europeans have a stronger preference for short working hours than Americans?⁴
- Is this due to "culture"? If so, what is culture?
- Attempts to reconcile micro and macro estimates of labor supply elasticities by considering a social multiplier: the utility of working less increases as others also work less⁵

⁴Blanchard, O. (2004), "The Economic Future of Europe", *Journal of Economic Perspectives*, 18, pp. 3-26

⁵Alesina, A., Glaeser, E.L. and Sacerdote, B. (2005), "Work and Leisure in the U.S. and Europe: Why So Different?", in *NBER Macroeconomics Annual 2005*, Vol. 20

Labor regulation and unions?

- Differences in labor law/regulation
- Vacation time
- Unions and collective agreements

The labor market

Labor supply and labor demand

Labor demand

- Firms choose labor input (conditional on wage rate) to maximize profits
- Jones ch. 4: demand decreasing in the wage because of decreasing marginal product of labor

Labor supply

- Households choose hours worked (conditional on wage rate) to maximize utility
- Income and substitution effects
- Long-run: evidence indicate that labor supply falls slightly as we become richer
- Substitution effect dominates in the shorter run, labor supply curve slopes up

Labor supply and labor demand

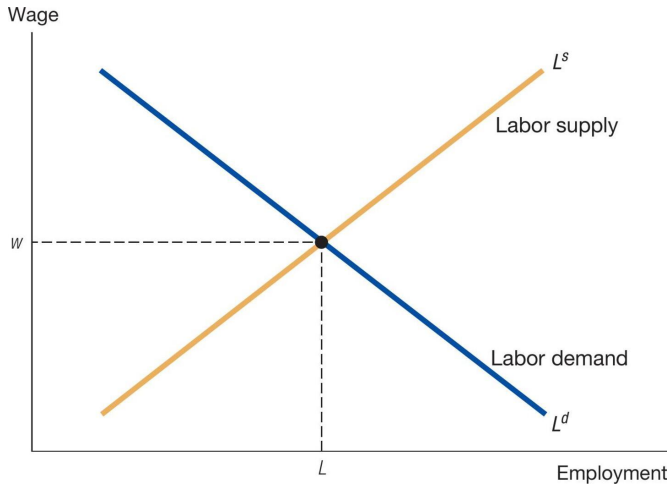


Figure 7.3 in Jones (2024)

Labor force, employment and unemployment

Employed: (a) persons who during the week worked at least 1 hour for pay or profit; (b) persons who were temporarily not at work because of holidays, sick leave, parental leave, etc.

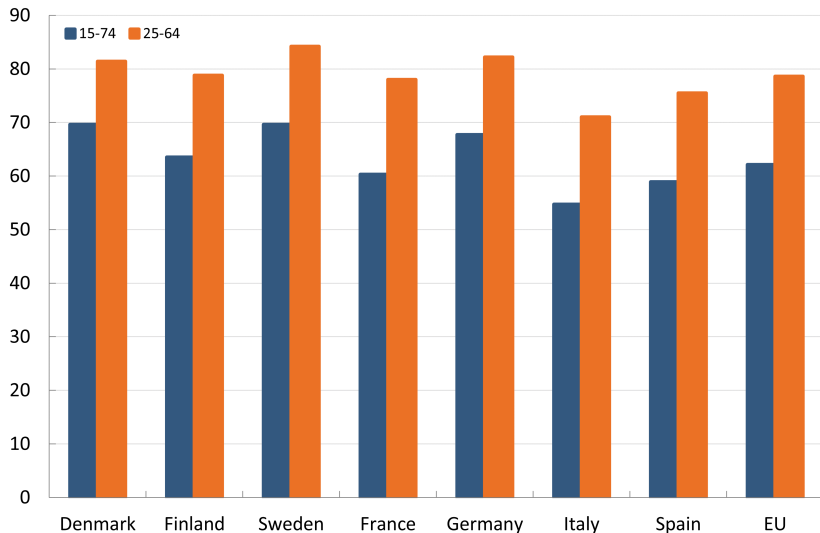
Unemployed: persons who were not employed, but who were available for work and actively searching for work

Labor force: Persons who are either employed or unemployed

Employment rate: Number of employed as a fraction of the number of persons in working age

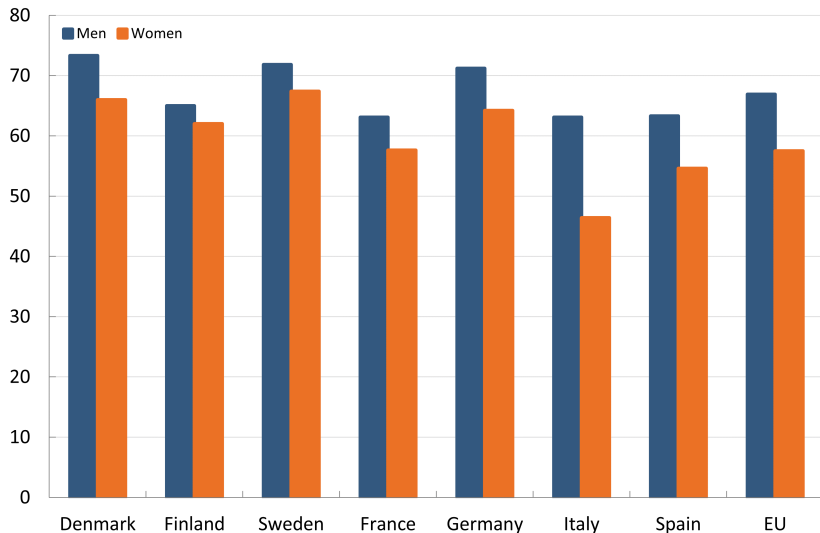
Unemployment rate: Number of unemployed as fraction of the labor force

Employment rates



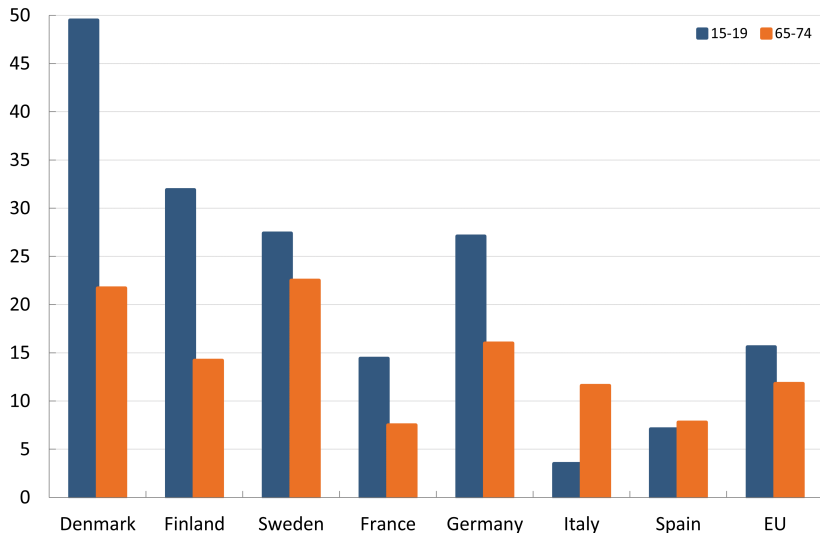
Employment rates in 2025Q2 for ages 15-74 and 25-64. Source: Eurostat LFS.

Employment rates for men and women



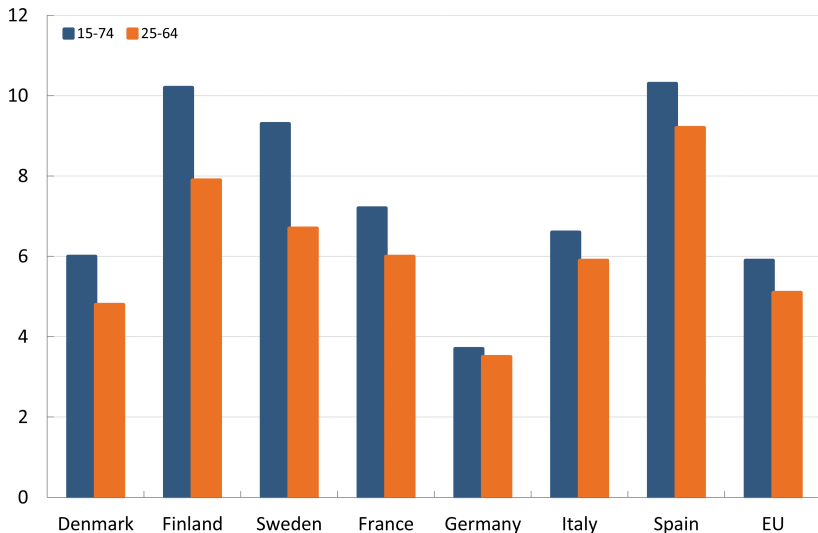
Employment rates in 2025Q2 for ages 15-74. Source: Eurostat LFS.

Employment rates for young and old



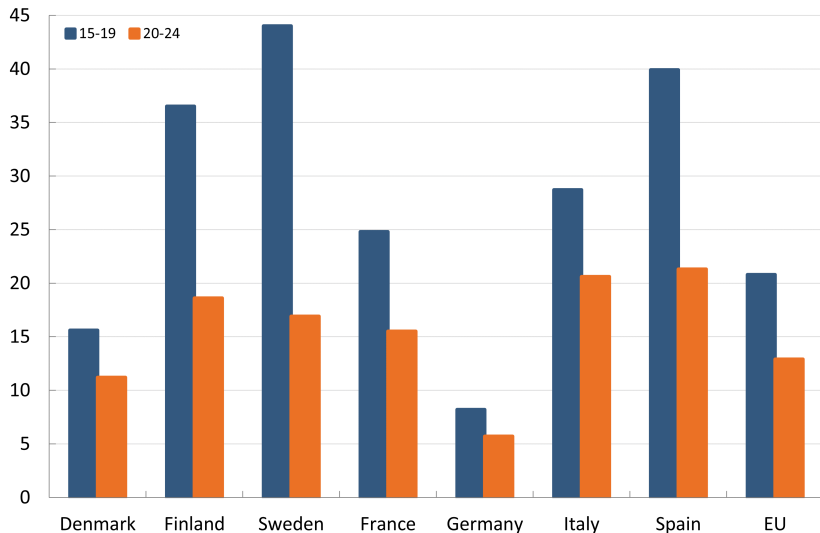
Employment rates in 2025Q2 for ages 15-19 and 65-74. Source: Eurostat LFS.

Unemployment rates



Unemployment rates in 2025Q2 for ages 15-74 and 25-64. Source: Eurostat LFS.

Youth unemployment



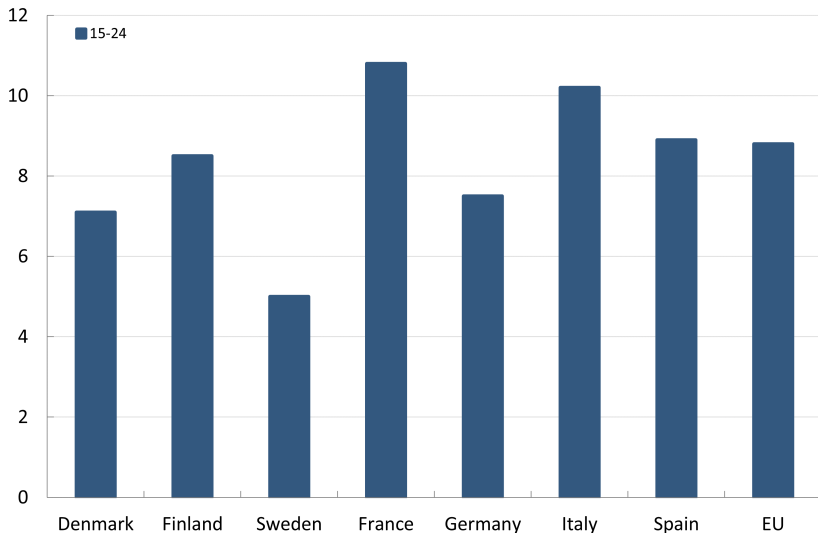
Unemployment rates in 2025Q2 for ages 15-19 and 20-24. Source: Eurostat LFS.

- Student looking for a part-time job will be counted as unemployed

Importance of such behavior varies a lot between countries and over the year

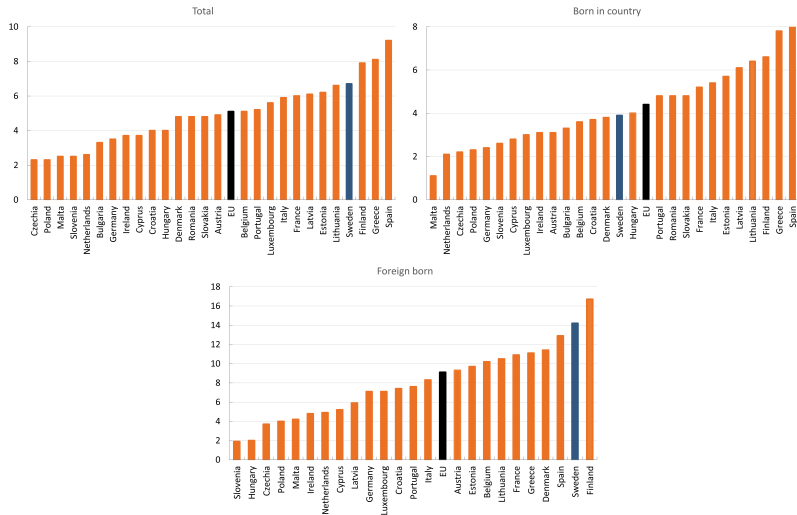
- Sweden: both the employment rate and the unemployment rate are high, in particular for the young
- *NEET*: people neither in employment nor in education or training
- Sweden: NEET rates are low

Youth inactivity



NEET rates in 2025Q2 for ages 15-24. Source: Eurostat LFS.

Unemployment by country of birth (ages 25-64)



Unemployment

Some unemployment concepts

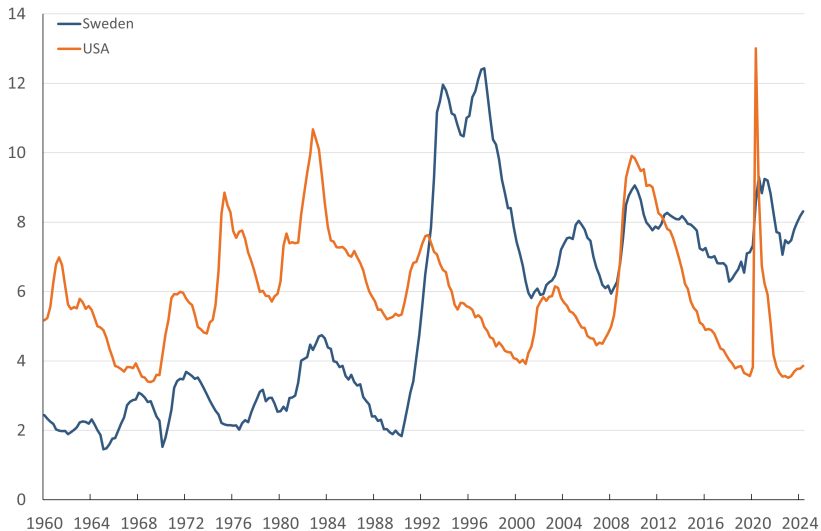
Natural rate of unemployment: the hypothetical unemployment rate that prevails if the economy is neither in a boom nor in a recession

Frictional unemployment: unemployment due to workers transitioning between jobs

Structural unemployment: unemployment due to labor market institutions and mismatch on the labor market

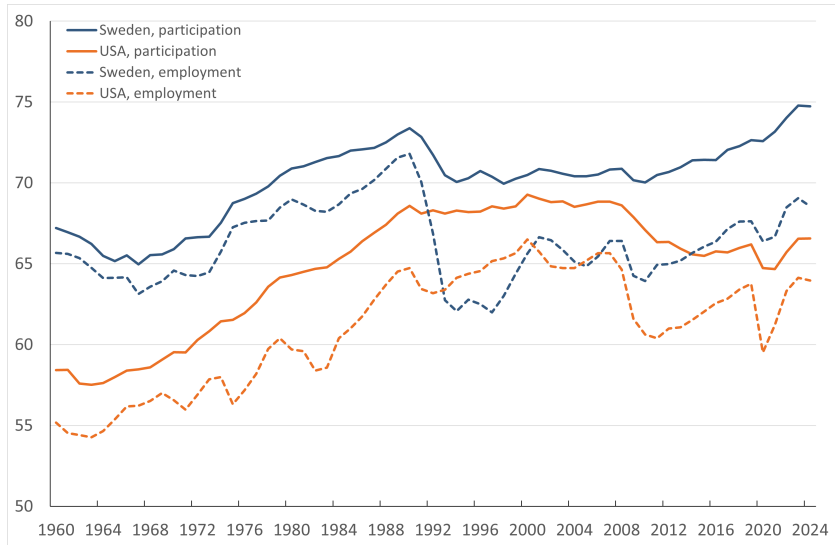
Cyclical unemployment: unemployment due to the business cycle (actual minus natural unemployment)

Unemployment over time



Unemployment rate rate for ages 15-74. Source: OECD Economic Outlook.

Labor force participation and employment over time



Labor-force participation rate and employment rate for ages 15-74. Source: OECD Economic Outlook.

Unemployment over time

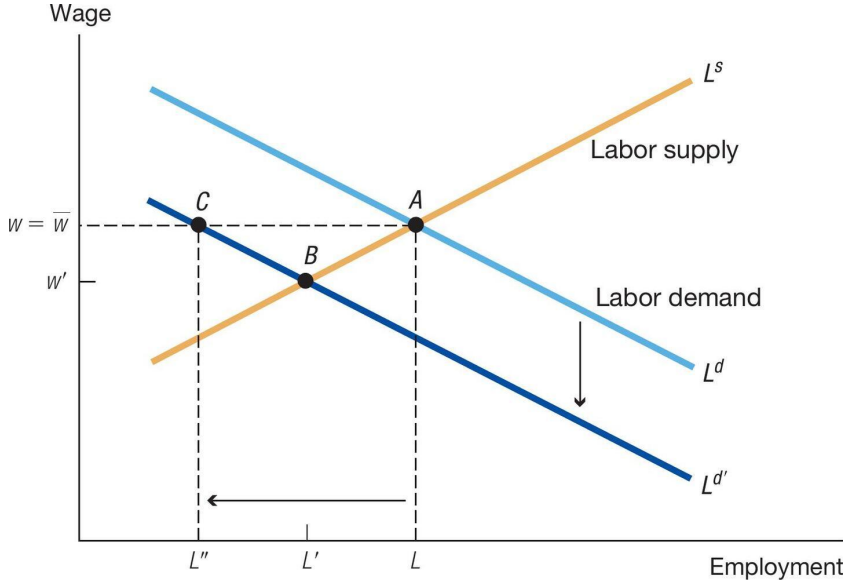
U.S. unemployment is volatile, but little trend

Swedish unemployment much lower prior to 1990s. Why?

Why did U.S. unemployment rise much more than Swedish in 2020?

Why is the labor-force participation rate so high in Sweden?

A reduction in labor demand with wage rigidity



The bathtub model of unemployment

A simple model of labor-market dynamics, with focus on natural rate of unemployment

Notation:

L : the labor force

E_t : the number of people employed at time t

U_t : the number of unemployed at time t

s : the job-separation rate

f : the job-finding rate

The bathtub model of unemployment

The labor force consists of employed and unemployed:

$$L = E_t + U_t$$

Flows on the labor market:

$$\Delta U_{t+1} \equiv U_{t+1} - U_t = sE_t - fU_t$$

The bathtub model of unemployment

Steady state when $\Delta U_{t+1} = 0$

Flows in and out of unemployment are then equal:

$$sE_t = fU_t$$

Use $E_t = L - U_t$ to find the steady-state unemployment rate:

$$s(L - U_t) = fU_t \implies u^* \equiv \frac{U^*}{L} = \frac{s}{f + s}$$

The bathtub model of unemployment

Shows how the natural rate of unemployment is affected by:

- separation rate
- finding rate

How are the separation and finding rates determined?

What labor market policies are effective?